

Group #5

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Project Overview

Understanding the relationship between environmental impact, economic development, and demographic shifts is vital for addressing global sustainability challenges. This project analyzes comprehensive datasets covering:

- CO₂ emissions per capita
- GDP per capita
- Population
- Continental grouping

The objective is to identify significant patterns and trends through data cleaning, integration, restructuring, and visual exploration. Using available data, this study investigates how various countries and continents compare and change over time.

How Project Meets ENSF 692 Requirements

- Designed and implemented a terminal-based data analysis application in Python.
- Imported and merged multiple datasets (CO₂, GDP, population, continents) using pandas.
- Original dataset includes 195 rows and 223 columns.
- Used two or more merge operations and removed duplicate columns/rows.
- Created a hierarchical index and performed multi-level sorting.
- Program includes user input for country and year selections, using try/except and validation logic.
- Created plots using matplotlib (scatter, line, bar, bubble).
- Used describe(), groupby(), pivot tables, and aggregation functions.
- Added calculated columns such as total GDP and CO₂ per GDP.
- Created custom functions for plotting, correlation analysis, and filtering.
- Exported cleaned dataset to Excel.

- Followed Python best practices and included docstrings for all functions and classes.
- Collaborated effectively via GitHub, and participated in the final presentation.

Dataset Citation

- CO₂ Emissions

Source: Gapminder, "Download the Data," Gapminder.org, 2025.
<https://www.gapminder.org/data/>

- GDP per Capita

Source: Gapminder, "Download the Data," Gapminder.org, 2025.
<https://www.gapminder.org/data/>

- Population

Source: Gapminder, "Download the Data," Gapminder.org, 2025.
<https://www.gapminder.org/data/>

- Country–Continent Mapping

Source: Kaggle, "Datasets | Kaggle," Kaggle.com, 2025.
<https://www.kaggle.com/datasets>